

Target-Aware and Scale-Adaptive Distributional Conformal Prediction for Stochastic Simulation Outputs

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Abstract

Predictive inference for stochastic simulation must provide reliable coverage for future simulation outputs while remaining informative under heteroscedastic and non-Gaussian response distributions. Distributional conformal prediction (DCP) is appealing because it calibrates a score based on an estimated conditional cumulative distribution function (CDF), but the coverage statement is only meaningful after specifying the input distribution over which future simulation scenarios are evaluated. We formulate simulation-output prediction relative to a user-specified target input distribution q_X , equivalently the target joint law $P_{XY}^q(dx, dy) = q_X(dx)P_{Y|X=x}(dy)$. This notation unifies ordinary global-domain coverage, region-of-interest coverage, and fixed-target-set average coverage. For the matched target-calibration setting, split conformal prediction gives finite-sample marginal validity under P_{XY}^q . To obtain efficient intervals under heterogeneous simulation noise, we use conditional kernel mean embeddings (CKME) to estimate the full conditional output law through smoothed indicator functions and introduce a scale-adaptive bandwidth $h(x) = c\hat{s}(x)$, where $\hat{s}(x)$ estimates the local response scale from training replications. Under a heteroscedastic location-scale model, oracle adaptive smoothing $h(x) = cs(x)$ removes the x -dependent standardized smoothing distortion created by a fixed bandwidth and restores homogeneity of the smoothed DCP score.

1 Introduction

Stochastic simulation is routinely used to evaluate complex systems whose outputs are noisy, heteroscedastic, skewed, or heavy-tailed. In such settings, a predictive interval should answer a distributional question: for a future run at input x , what range of outputs Y should be expected with a prescribed probability? This question is not only about mean prediction. It also requires local information about the conditional response distribution $P_{Y|X=x}$, because simulation variability can change sharply across the input space.

Conformal prediction (CP) provides finite-sample, distribution-free coverage guarantees under exchangeability [13, 10, 6, 1]. Standard residual-based CP, however, calibrates a global residual scale. Under heteroscedasticity this can produce intervals that are too wide in low-variance regions and too narrow or unstable in high-variance regions. Distributional conformal prediction (DCP) addresses this issue by constructing conformity scores from an estimated conditional CDF [4]. If the conditional CDF is accurate, the probability integral transform maps responses onto a common probability scale, making interval construction more adaptive to local distributional shape.

For simulation output analysis, two issues remain underdeveloped. First, the coverage statement must specify the *target input distribution*. Simulation users often do not care about a natural

population distribution over inputs; they may care about a design distribution over the whole experimental domain, a high-risk or policy-relevant region of interest, or a finite set of planned target scenarios. We denote this input target by q_X . The corresponding joint law is

$$P_{XY}^q(dx, dy) = q_X(dx)P_{Y|X=x}(dy),$$

and the intended coverage statement is marginal under this target joint law. Split conformal validity applies cleanly only when calibration inputs and target inputs are exchangeable under the same q_X . If calibration points are drawn from one design distribution and target points from another, the usual guarantee no longer follows without additional machinery such as weighted conformal calibration.

Second, when CKME is used to estimate a conditional CDF, the CDF is obtained by evaluating smoothed indicator functions. The smoothing bandwidth is not a harmless numerical detail. With a fixed bandwidth h , the standardized amount of smoothing is $h/s(x)$, where $s(x)$ is the local response scale. Thus the same scalar h oversmooths low-scale regions and undersmooths high-scale regions. This creates an avoidable distortion in the estimated DCP score, even if the conformal calibration step still protects marginal coverage.

This paper reframes CKME-DCP around these two principles: validity is target-distribution dependent, and efficiency is score-scale dependent. We use CKME as the conditional distribution estimator because it estimates a full conditional law as an RKHS-valued object and evaluates conditional CDFs through smooth test functions. This representation exposes a natural scale-adaptive smoothing rule,

$$h(x) = c\hat{s}(x),$$

where $\hat{s}(x)$ is estimated from training-stage simulation replications. The result is a conformal procedure that keeps the finite-sample split-CP guarantee at the target-distribution level while improving conditional efficiency through better local CDF smoothing.

The contributions are as follows.

1. We formulate CKME-DCP under a target joint law $P_{XY}^q(dx, dy) = q_X(dx)P_{Y|X=x}(dy)$. This makes clear whether coverage is claimed globally, over a region of interest, or on average over a fixed target set.
2. We introduce scale-adaptive CKME-DCP. It replaces a single smooth-indicator bandwidth with $h(x) = c\hat{s}(x)$, where $\hat{s}(x)$ is learned from training replications and smoothed across input space.
3. We separate validity from efficiency. Split conformal validity is estimator-agnostic and follows when calibration and target samples are exchangeable under q_X . Uniform CDF error transfers to DCP score and threshold error, affecting interval efficiency and group-conditional behavior.
4. We show that, under a heteroscedastic location-scale model, oracle adaptive smoothing $h(x) = cs(x)$ removes the x -dependent standardized smoothing ratio $h/s(x)$ induced by fixed smoothing and restores homogeneity of the smoothed-oracle DCP score.

2 Related Work

2.1 Conformal prediction and distributional scores

Conformal prediction constructs finite-sample prediction sets by calibrating a user-chosen non-conformity score on exchangeable data [13, 10]. Split conformal prediction separates model fitting

from calibration and therefore applies to arbitrary predictive models, including black-box regression, Gaussian-process metamodels, distribution regression, and kernel-based conditional distribution estimators [6, 1]. This estimator-agnostic validity is central for simulation output analysis because the stochastic simulator may be expensive, nonlinear, and misspecified relative to any parametric metamodel.

The choice of score determines the geometry and efficiency of the resulting prediction set. Residual scores are simple and robust, but they often behave like a global-scale correction. Conformalized quantile regression uses estimated conditional quantiles to construct locally adaptive intervals [8]. Distributional conformal prediction instead estimates a conditional CDF and calibrates a score on the probability scale [4]. The present paper follows the distributional route because simulation analysts often need more than two quantiles: they may want the full conditional output distribution, interval inversion at multiple confidence levels, tail probabilities, or service-level probabilities. CKME is useful in this setting because it represents the conditional law as a single object rather than as a separate regression for every threshold or quantile.

2.2 Target, local, and weighted conformal inference

Conformal validity is a marginal statement over the distribution that generates the calibration and future examples. Recent work on weighted conformal prediction, covariate shift, and localized conformal prediction makes this dependence explicit by modifying calibration weights or neighborhoods when the target covariate distribution differs from the calibration distribution [12, 5]. Our target notation q_X is aligned with this literature, but our main operating regime is deliberately cleaner: the calibration and evaluation inputs are sampled from the same target distribution. This lets the paper keep the finite-sample theorem line simple while still making the scope of the guarantee visible.

This distinction is important in simulation. In observational prediction, the test covariate distribution may be viewed as given by nature. In simulation, the analyst usually chooses the design points, calibration points, and evaluation scenarios. A statement such as “90% coverage” is therefore incomplete unless it says whether the probability is over the whole design region, a safety-critical region of interest, a traffic peak, or a fixed portfolio of decision scenarios. Our contribution is not to solve every target-distribution mismatch problem. Rather, it is to put the target distribution into the formulation before introducing CKME-DCP and to separate matched-target validity from optional reweighting or local-calibration extensions.

2.3 Simulation metamodeling and uncertainty quantification

Classical simulation metamodeling focuses on learning the response surface of a stochastic simulator from designed experiments. Gaussian-process and stochastic-kriging methods are widely used for mean prediction and uncertainty quantification under simulation noise [9, 2]. Heteroscedastic Gaussian-process models further allow the noise variance to vary across the design space and are effective for many replication-based simulation settings [3]. These methods provide model-based predictive distributions or credible intervals, but their nominal uncertainty statements depend on the modeling assumptions.

Conformal calibration provides a complementary layer: it can wrap a fitted simulator metamodel or conditional distribution estimator and supply finite-sample marginal coverage under exchangeability. The price is that conformal calibration alone does not guarantee local efficiency. A poor score can still yield intervals that are too conservative in some regions and too aggressive in others. This paper therefore treats CKME and scale-adaptive smoothing as the score-construction layer,

while split conformal calibration remains the validity layer.

2.4 Kernel embeddings for conditional distributions

Kernel mean embeddings represent probability distributions as elements of a reproducing kernel Hilbert space, and conditional kernel mean embeddings extend this representation to conditional laws [11, 7]. In the present application, the CKME representation is attractive because DCP needs repeated evaluations of $F(t | x)$ over a grid of candidate response values. Once the CKME weights are computed, the conditional expectation of a smooth test function can be evaluated by a weighted sum over training outputs.

The key technical issue is that the CDF indicator $1\{Y \leq t\}$ is typically not a smooth RKHS function. CKME-DCP therefore uses smooth indicator functions. This creates a bandwidth parameter on the output scale. Previous CKME-DCP use treats this bandwidth as a scalar tuning parameter. The present paper shows why that scalar choice is problematic under heteroscedasticity and proposes the scale-adaptive rule $h(x) = c\hat{s}(x)$ as a direct correction to the standardized smoothing distortion.

3 Target-Aware Predictive Inference

3.1 Target input distributions and target joint laws

Let $X \in \mathcal{X} \subseteq \mathbb{R}^d$ denote a simulation input and $Y \in \mathbb{R}$ the corresponding stochastic output. We write $P_{Y|X=x}$ for the conditional output law at input x . Unlike many supervised-learning problems, the input distribution in simulation is often chosen by the analyst. We therefore distinguish three input distributions:

$$p_{\text{tr}}(x), \quad p_{\text{cal}}(x), \quad q_X(x),$$

where p_{tr} is the training design distribution used to fit the conditional CDF estimator, p_{cal} is the calibration design distribution, and q_X is the target distribution over which coverage is claimed.

Definition 1 (Target joint law and target coverage). *For a user-specified target input distribution q_X , define*

$$P_{XY}^q(dx, dy) = q_X(dx)P_{Y|X=x}(dy).$$

A prediction set $\hat{C}$ has marginal target coverage $1 - \alpha$ if

$$\mathbb{P}_{(X,Y) \sim P_{XY}^q} \{Y \in \hat{C}(X)\} \geq 1 - \alpha.$$

This is not a new type of conformal validity; it is the standard marginal conformal statement evaluated under a user-specified target joint law. Different choices of q_X recover different simulation goals:

- *Global-domain target:* $q_X = P_X$, or a user-chosen space-filling distribution over the simulation domain, gives ordinary marginal coverage over that domain.
- *Region-of-interest target:* $q_X = P_X(\cdot | X \in \mathcal{R})$, or another distribution concentrated on $\mathcal{R}$, gives

$$\mathbb{P}\{Y \in \hat{C}(X) | X \in \mathcal{R}\} \geq 1 - \alpha.$$

This is event-conditional or group coverage for the region $\mathcal{R}$, not pointwise conditional coverage at every $x \in \mathcal{R}$.

- *Fixed-target-set average*: for planned simulation scenarios $x_1^*, \dots, x_m^*$, the discrete target

$$q_X = m^{-1} \sum_{j=1}^m \delta_{x_j^*}$$

corresponds to the average scenario guarantee

$$\frac{1}{m} \sum_{j=1}^m \mathbb{P}_{Y|X=x_j^*} \{Y \in \widehat{C}(x_j^*)\} \geq 1 - \alpha.$$

This does not guarantee coverage at each individual scenario.

- *Smoothed fixed-target neighborhoods*: one may also define

$$q_\tau(x) = m^{-1} \sum_{j=1}^m K_\tau(x - x_j^*)$$

and report a local guarantee around the target set rather than exactly on the discrete set.

This target notation prevents a common ambiguity. If calibration inputs are sampled globally but evaluation is restricted to a region of interest, the calibration and target pairs are generally not exchangeable. The usual split-conformal guarantee does not automatically transfer to that new target.

3.2 Matched target calibration as a design requirement

The notation $p_{\text{tr}}, p_{\text{cal}}, q_X$ also clarifies the roles of the different data sources in a simulation study. The training design p_{tr} controls where simulation budget is spent to learn the conditional distribution. It may be space-filling, adaptive, or concentrated where the simulator is difficult. The calibration distribution p_{cal} , however, determines the distribution of conformal scores used to set the prediction threshold. The evaluation or deployment target is q_X . For the standard split-conformal proof to apply without weighting, the calibration pairs and the future target pair must be exchangeable after conditioning on the fitted estimator.

This means that p_{tr} can differ from q_X without invalidating split conformal calibration, provided that the fitted estimator is treated as fixed during calibration and the calibration sample itself follows q_X . Such a separation is natural in simulation. One may spend the training budget broadly to learn the response surface, then spend the calibration budget on the operational region where guarantees are needed. Conversely, if calibration is global but evaluation is restricted to a high-risk region, the empirical coverage estimate is a diagnostic for that region, not a finite-sample conformal guarantee for that region.

The matched target-calibration regime is the main regime of this paper. It leads to the clean theorem line:

$$(X_i^{\text{cal}}, Y_i^{\text{cal}}), (X_\star, Y_\star) \stackrel{\text{exch.}}{\sim} P_{XY}^q \implies \mathbb{P}\{Y_\star \in \widehat{C}(X_\star)\} \geq 1 - \alpha.$$

The statement is intentionally marginal under q_X . It is not a pointwise conditional guarantee, and it does not claim validity for target distributions that were not used in calibration. This boundary also explains why bin-wise coverage, when reported, should be interpreted as a diagnostic rather than as a separate finite-sample guarantee.

Remark 1 (Why keep p_{tr} separate?). If p_{tr} is too far from q_X , the CKME estimator may be inaccurate in the target region, producing wide or unstable intervals after calibration. This is an efficiency and estimation problem, not a conformal validity problem, as long as the calibration and target pairs are exchangeable. In practice, the training design should cover the support of q_X well enough for conditional distribution estimation, while the calibration design should match q_X for validity.

3.3 Split conformal prediction and DCP scores

Let $Z = (X, Y) \in \mathcal{X} \times \mathcal{Y}$ denote an input-output pair drawn from the target joint law P_{XY}^q . For a new target input $X_\star$, we seek a set-valued predictor $\widehat{C}$ such that $Y_\star \in \widehat{C}(X_\star)$ with probability at least $1 - \alpha$. Conformal prediction provides such marginal coverage under exchangeability: conditional on any fitted model or score construction, the calibration pairs and the target pair must have a joint distribution invariant to permutations.

The construction relies on a nonconformity score $S : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$, where larger values indicate that a candidate pair (x, y) is less compatible with the fitted model. A common regression score is the absolute residual $S(x, y) = |y - \widehat{\mu}(x)|$, but the conformal argument does not depend on this particular choice. Split conformal prediction separates model fitting from calibration. A training sample $\mathcal{D}_{\text{tr}}$ is used to fit the predictive model or distribution estimator, and an independent calibration sample $\mathcal{D}_{\text{cal}}$ is used only to calibrate the score. For a fitted score function $S(x, y)$, the calibration scores are

$$S_i^{\text{cal}} = S(X_i^{\text{cal}}, Y_i^{\text{cal}}), \quad i = 1, \dots, n_{\text{cal}}.$$

Let $\widehat{q}_{1-\alpha}$ be the $\lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil$ -th order statistic of these scores. The split conformal prediction set is

$$\widehat{C}(x) = \{y : S(x, y) \leq \widehat{q}_{1-\alpha}\}.$$

Conditional on the fitted score, exchangeability of the calibration pair and the target pair is enough to obtain finite-sample marginal coverage. The score can be residual-based, quantile-based, localized, or distributional; the validity argument is the same.

Standard residual-based CP can be inefficient under heteroscedasticity because a single calibrated residual threshold tends to impose a nearly global response-scale correction. Distributional conformal prediction addresses this by using a score based on an estimated conditional CDF. Let

$$F(t \mid x) = \mathbb{P}(Y \leq t \mid X = x)$$

denote the conditional CDF. Assume for motivation that $F(\cdot \mid x)$ is continuous for almost every x . Given an estimator $\widehat{F}(\cdot \mid x)$ fitted on $\mathcal{D}_{\text{tr}}$, DCP defines a distributional nonconformity score at a candidate point (x, y) . A common choice, and the one used in this paper, is

$$S_{\widehat{F}}(x, y) = \left| \widehat{F}(y \mid x) - 1/2 \right|. \quad (1)$$

More generally one may use $A(\widehat{F}(y \mid x))$ for a suitable transformation A . The score in (1) is motivated by the probability integral transform: if the true continuous CDF is known, then $F(Y \mid X)$ is uniformly distributed on $[0, 1]$ and no longer carries the original response scale. Thus DCP measures nonconformity on a probability scale rather than directly on the response scale, allowing prediction sets to adapt to heteroscedasticity, skewness, and tail behavior through the estimated conditional distribution.

Applying split conformal calibration to the DCP score yields

$$\widehat{C}_{\text{DCP}}(x) = \{y \in \mathcal{Y} : S_{\widehat{F}}(x, y) \leq \widehat{q}_{1-\alpha}\}.$$

The conditional CDF estimator $\hat{F}$ can be obtained by distribution regression, quantile-based methods, or other flexible conditional distribution estimators. Its accuracy primarily affects sharpness, interval shape, and groupwise behavior, while split conformal calibration supplies marginal validity under the target law. This is why the present paper separates the target-distribution validity layer from the CKME estimation and scale-adaptive smoothing layer.

3.4 Split conformal validity under a target distribution

Let $\mathcal{D}_{\text{tr}}$ be a training sample used to fit a conditional CDF estimator $\hat{F}$. Conditional on $\mathcal{D}_{\text{tr}}$, let

$$(X_i^{\text{cal}}, Y_i^{\text{cal}}), \quad i = 1, \dots, n_{\text{cal}},$$

be independent calibration pairs from P_{XY}^q , i.e., $X_i^{\text{cal}} \sim q_X$ and $Y_i^{\text{cal}} \sim P_{Y|X=X_i^{\text{cal}}}$. For a candidate pair (x, y) , use the DCP score

$$S_{\hat{F}}(x, y) = \left| \hat{F}(y | x) - 1/2 \right|. \quad (2)$$

Let $\hat{q}_{1-\alpha}$ be the $[(n_{\text{cal}} + 1)(1 - \alpha)]$ -th order statistic of the calibration scores. The split DCP prediction set is

$$\hat{C}(x) = \{y \in \mathbb{R} : S_{\hat{F}}(x, y) \leq \hat{q}_{1-\alpha}\}. \quad (3)$$

Proposition 1 (Target-distribution split conformal validity). *Suppose that the calibration pairs and the target pair are exchangeable conditional on the fitted estimator $\hat{F}$. In particular, this holds when calibration pairs and the target pair are independently drawn from the same target joint law P_{XY}^q . Then the prediction set in (3) satisfies*

$$\mathbb{P}_{(X,Y) \sim P_{XY}^q} \{Y \in \hat{C}(X)\} \geq 1 - \alpha.$$

This guarantee does not require $\hat{F}$ to be correctly specified.

Proposition 1 states the validity line of the paper. CKME and adaptive smoothing are not needed for marginal conformal validity. They affect the quality, shape, and conditional efficiency of the resulting intervals.

Remark 2 (When calibration and target distributions differ). *If calibration inputs follow p_{cal} but target inputs follow q_X , unweighted split conformal calibration is generally calibrated for p_{cal} , not for q_X . Under support overlap, weighted conformal calibration can be used with weights proportional to $q_X(x)/p_{\text{cal}}(x)$ [12]. This paper treats the weighted version as part of the target-aware formulation, while the main theorem line focuses on matched calibration and target distributions.*

Remark 3 (Fixed target inputs). *A deterministic list of fixed target inputs is not automatically an exchangeable test sample from a global calibration design. If calibration and evaluation are both organized around the discrete target $m^{-1} \sum_j \delta_{x_j^*}$, the conformal claim is an average over the target scenarios, not a separate guarantee for every x_j^* . If nearby calibration points are more natural than exact replication at the targets, a practical alternative is to smooth the target set into*

$$q_\tau(x) = m^{-1} \sum_{j=1}^m K_\tau(x - x_j^*),$$

collect or reweight calibration data for q_τ , and report coverage for that local target distribution. This is a target-distribution issue, not a bandwidth issue in the CKME CDF estimator.

4 Scale-Adaptive CKME-DCP

4.1 CKME conditional CDF estimation

Conditional kernel mean embedding provides a nonparametric representation of the full conditional output law. Rather than fitting a separate model for each threshold t , CKME embeds $P_{Y|X=x}$ as a single RKHS-valued object and evaluates conditional expectations through inner products [11, 7]. This is useful for DCP because the score requires the conditional CDF over many candidate response values when inverting the prediction set.

Let $\mathcal{H}_X$ and $\mathcal{H}_Y$ be RKHSs with kernels k_X on $\mathcal{X}$ and k_Y on $\mathcal{Y}$. We assume these kernels are bounded, e.g.,

$$\sup_{x \in \mathcal{X}} k_X(x, x) \leq \kappa_X^2, \quad \sup_{y \in \mathcal{Y}} k_Y(y, y) \leq \kappa_Y^2.$$

Let $\phi_Y : \mathcal{Y} \rightarrow \mathcal{H}_Y$ denote the output feature map. Analogous to a kernel mean embedding for a marginal distribution, CKME represents the conditional distribution $P_{Y|X=x}$ by the conditional mean embedding

$$\mu_{Y|x} = \mathbb{E}[\phi_Y(Y) | X = x] \in \mathcal{H}_Y.$$

This representation is useful because, for any $g \in \mathcal{H}_Y$, the conditional expectation satisfies

$$\mathbb{E}[g(Y) | X = x] = \langle \mu_{Y|x}, g \rangle_{\mathcal{H}_Y}.$$

Hence CKME provides a compact distributional representation from which many conditional expectations can be recovered.

For conditional CDF estimation, $F(t | x) = \mathbb{P}(Y \leq t | X = x)$ would correspond to $g(y) = \mathbf{1}\{y \leq t\}$. Ideally one would write $F(t | x) = \langle \mu_{Y|x}, \mathbf{1}\{\cdot \leq t\} \rangle_{\mathcal{H}_Y}$. However, this identity requires the indicator function to belong to $\mathcal{H}_Y$, which generally fails for smooth output kernels such as Gaussian or Matérn kernels. We therefore replace the hard indicator with a smooth surrogate.

Let $\Psi : \mathbb{R} \rightarrow [0, 1]$ be nondecreasing and Lipschitz, approximating $\mathbf{1}\{u \geq 0\}$. For bandwidth $h > 0$, define

$$g_{t,h}(y) = \Psi\left(\frac{t - y}{h}\right).$$

The corresponding smoothed conditional CDF is

$$F_h(t | x) = \mathbb{E}[g_{t,h}(Y) | X = x] = \langle \mu_{Y|x}, g_{t,h} \rangle_{\mathcal{H}_Y},$$

provided $g_{t,h} \in \mathcal{H}_Y$. The bandwidth h therefore controls output-side smoothing in the CDF estimator.

Given training data $\mathcal{D}_{\text{tr}} = \{(X_i, Y_i)\}_{i=1}^{n_{\text{tr}}}$, CKME can be viewed as vector-valued kernel ridge regression for the conditional embedding. The fitted embedding has the closed form

$$\hat{\mu}_{Y|x} = \sum_{i=1}^{n_{\text{tr}}} a_i(x) \phi_Y(Y_i),$$

with weights

$$a(x) = (K_X + n_{\text{tr}} \lambda I)^{-1} k_X(x),$$

where K_X is the $n_{\text{tr}} \times n_{\text{tr}}$ input Gram matrix with entries $[K_X]_{ij} = k_X(X_i, X_j)$, and

$$k_X(x) = (k_X(X_1, x), \dots, k_X(X_{n_{\text{tr}}}, x))^{\top}.$$

Here $\lambda > 0$ is the regularization parameter and I is the identity matrix. Substituting $\hat{\mu}_{Y|x}$ into the smoothed CDF representation gives

$$\hat{F}_h(t | x) = \langle \hat{\mu}_{Y|x}, g_{t,h} \rangle_{\mathcal{H}_Y} = \sum_{i=1}^{n_{tr}} a_i(x) \Psi\left(\frac{t - Y_i}{h}\right). \quad (4)$$

With an input-dependent bandwidth, we evaluate

$$\hat{F}_{h(\cdot)}(t | x) = \sum_{i=1}^{n_{tr}} a_i(x) \Psi\left(\frac{t - Y_i}{h(x)}\right). \quad (5)$$

The CKME weights $a(x)$ depend on the input kernel and regularization, not on the smoothing bandwidth. Thus $h(x)$ can be changed at prediction and calibration time without retraining the CKME coefficient system.

Remark 4 (Why not use hard empirical indicators?). *One can estimate $F(t | x)$ by replacing $g_{t,h}(Y_i)$ in (4) with $\mathbf{1}\{Y_i \leq t\}$. For each fixed t , this is a scalar kernel ridge regression problem for the binary response $\mathbf{1}\{Y \leq t\}$. It avoids smoothing bias, but it is no longer the CKME inner-product estimator in $\mathcal{H}_Y$, and it removes the bandwidth through which local response scale is controlled. Since the contribution of this paper is scale-adaptive score construction through $h(x)$, we use smooth indicators as the main estimator. Hard indicators are best treated as an ablation for calibrated interval performance, not as the main empirical race in conditional CDF estimation.*

4.2 Local scale and adaptive bandwidth

Here *scale* means the local spread of the simulation output distribution $Y | X = x$. In a location-scale model this is

$$Y = m(X) + s(X)\varepsilon, \quad \varepsilon \perp X.$$

For Gaussian outputs, $s(x)$ is the conditional standard deviation; for Student- t outputs, it is the scale parameter multiplying the standardized noise. The fixed-bandwidth estimator uses the same response-scale smoothing everywhere. In standardized units, however, the effective bandwidth is

$$h_{\text{eff}}(x) = \frac{h}{s(x)}.$$

Thus a fixed h can oversmooth low-noise regions, where $h/s(x)$ is large, and undersmooth high-noise regions, where $h/s(x)$ is small. This motivates the adaptive rule

$$h(x) = c\hat{s}(x), \quad (6)$$

where $c > 0$ is a global multiplier and $\hat{s}(x)$ is estimated from training-stage simulation replications. The goal is to make $h(x)/s(x)$ approximately stable across inputs.

In the current implementation, the training data are collected at design sites x_i with r_0 replications per site. At each site we compute the robust scale estimate

$$\hat{s}_i = \frac{\text{IQR}(Y_{i1}, \dots, Y_{ir_0})}{1.349},$$

then smooth $(x_i, \hat{s}_i)$ over $\mathcal{X}$ using Nadaraya-Watson regression with a Gaussian kernel. The resulting $\hat{s}(x)$ is floored to avoid degenerate bandwidths, and $h(x) = c\hat{s}(x)$ is used both for calibration scores and for prediction-set inversion.

The IQR estimator is not the only possible scale estimator. A sample standard deviation is efficient under Gaussian noise, but it can be unstable when replications are heavy-tailed or when a small number of extreme simulation outputs occur. A median absolute deviation estimator is another robust option. We use IQR because it has a simple Gaussian calibration constant, is stable under heavy-tailed replications, and is easy to compute from site-level replications. Since split conformal calibration absorbs moderate multiplicative scale errors, the purpose of $\widehat{s}(x)$ is not to estimate $s(x)$ perfectly. Its purpose is to remove large systematic changes in the standardized smoothing ratio across the input space.

The multiplier c controls the global amount of smoothing after scale normalization. When c is too small, the smoothed indicator becomes nearly discontinuous and the CKME CDF estimate can be noisy. When c is too large, the CDF estimate is biased toward an overly diffuse distribution and the DCP score loses resolution. A small sweep over c can be used to check whether $c = 1$ is a stable default. A more elaborate implementation could tune c by cross-validation on a distributional loss, but we keep the default simple because the main contribution is the input-dependent scale normalization rather than another scalar tuning routine.

4.3 Prediction-set construction

The full scale-adaptive CKME-DCP procedure is:

1. Collect training data $\mathcal{D}_{\text{tr}}$ and fit CKME weights with tuned (ℓ_x, λ) .
2. Estimate the local scale function $\widehat{s}(x)$ from training replications and set $h(x) = c\widehat{s}(x)$.
3. For each calibration pair $(X_i^{\text{cal}}, Y_i^{\text{cal}})$, compute

$$\widehat{S}_i = \left| \widehat{F}_{h(\cdot)}(Y_i^{\text{cal}} \mid X_i^{\text{cal}}) - 1/2 \right|.$$

4. Let $\widehat{q}_{1-\alpha}$ be the split-conformal quantile of $\widehat{S}_1, \dots, \widehat{S}_{n_{\text{cal}}}$.
5. For a target input x , output

$$\widehat{C}(x) = \left\{ y \in \mathcal{T} : \left| \widehat{F}_{h(\cdot)}(y \mid x) - 1/2 \right| \leq \widehat{q}_{1-\alpha} \right\},$$

where $\mathcal{T}$ is the numerical grid used for inversion.

4.4 Implementation details

The implementation follows the training-calibration split used in the WSC version of CKME-DCP, with one change: the output-side bandwidth is evaluated as a function of the prediction input. The CKME hyperparameters (ℓ_x, λ) are selected using the training data. The fixed-bandwidth baseline also tunes a scalar h . The adaptive version replaces this scalar with $h(x) = c\widehat{s}(x)$, using the same CKME input weights $a(x)$. Thus the computationally expensive matrix solve for $(K_X + n_{\text{tr}}\lambda I)^{-1}$ is unchanged.

Prediction-set inversion is performed on a response grid $\mathcal{T}$. In one-dimensional output settings this is straightforward: evaluate the DCP score $S_{\widehat{F}}(x, t)$ for every $t \in \mathcal{T}$ and return the grid values whose scores are below the calibrated threshold. The resulting set may be an interval or a union of intervals when the estimated CDF is imperfect. For reporting one-dimensional prediction intervals, we summarize it by the smallest interval covering the accepted grid values. This convention matches the simulation-output setting where the user usually wants a one-dimensional prediction interval.

For repeated simulation outputs, there are two equivalent data views. One may treat each replication as an input-output pair (x_i, Y_{ij}) , which is convenient for CKME fitting, while also preserving the site structure to estimate $\hat{s}_i$ from the replications at each x_i . The local-scale estimator therefore uses the replication structure, but the conformal calibration scores are computed at the individual output level. The calibration sample is independent of the training sample used to fit CKME and estimate $\hat{s}$.

The method has three distinct tuning components. The input kernel length scale controls how information is shared across neighboring simulation inputs. The regularization parameter controls the stability of the CKME weights. The output bandwidth controls smooth-indicator resolution. The proposed adaptive rule changes only the third component. This separation is important for interpreting future ablations: improvement over fixed-bandwidth CKME-DCP would be evidence that the response-scale smoothing correction matters, rather than evidence that the entire CKME estimator has been replaced.

5 Theory

5.1 Estimator error and conformal behavior

Let $F^\circ(t \mid x)$ denote the oracle CDF target. Depending on context, this can be the exact conditional CDF or the smoothed oracle CDF corresponding to a chosen $h(x)$. Define the oracle score

$$S^\circ(x, y) = |F^\circ(y \mid x) - 1/2|.$$

All theoretical comparisons below are made over a response range $\mathcal{T}$ used for score evaluation and prediction-set inversion. We assume $\mathbb{P}(Y \in \mathcal{T}) = 1$ under the target law, or equivalently that $\mathcal{T}$ is chosen large enough that the omitted tail probability is negligible and handled outside the theorem.

Assumption 1 (Uniform CDF error). *There exists $\epsilon_n = o_{\mathbb{P}}(1)$ such that*

$$\sup_{x \in \mathcal{X}, t \in \mathcal{T}} \left| \hat{F}(t \mid x) - F^\circ(t \mid x) \right| \leq \epsilon_n$$

with probability tending to one.

Assumption 2 (Regular oracle score distribution). *The marginal distribution of $S^\circ(X, Y)$, with $X \sim q_X$, is continuous and has a density g° bounded away from zero and infinity in a neighborhood of its $(1 - \alpha)$ -quantile $q_{1-\alpha}^\circ$.*

Proposition 2 (CDF error transfers to DCP score and threshold error). *Under Assumptions 1–2,*

$$\sup_{x, y} |S_{\hat{F}}(x, y) - S^\circ(x, y)| \leq \epsilon_n$$

with probability tending to one, and the split-conformal threshold satisfies

$$|\hat{q}_{1-\alpha} - q_{1-\alpha}^\circ| = O_{\mathbb{P}}\left(\epsilon_n + n_{\text{cal}}^{-1/2}\right).$$

This proposition clarifies the role of the conditional CDF estimator. Finite-sample marginal validity comes from conformal calibration. CDF accuracy controls how close the calibrated prediction set is to an oracle DCP set and therefore affects width, interval score, and group-conditional coverage.

5.2 Why scale-adaptive smoothing helps

Assume the heteroscedastic location-scale model

$$Y = m(X) + s(X)\varepsilon, \quad \varepsilon \perp X, \quad s(x) > 0. \quad (7)$$

For fixed h , the smoothed oracle CDF is

$$F_h^\circ(t \mid x) = \mathbb{E} \left[\Psi \left(\frac{t - Y}{h} \right) \middle| X = x \right].$$

Writing $z = (t - m(x))/s(x)$, we have

$$F_h^\circ(t \mid x) = \mathbb{E} \left[\Psi \left(\frac{z - \varepsilon}{h/s(x)} \right) \right]. \quad (8)$$

Thus the standardized smoothing ratio $h/s(x)$ varies with x . A fixed h does not represent the same CDF resolution in low- and high-variance regions.

Now set $h(x) = cs(x)$. The smoothed oracle CDF becomes

$$F_c^\circ(t \mid x) = H_c \left(\frac{t - m(x)}{s(x)} \right), \quad H_c(z) = \mathbb{E} \left[\Psi \left(\frac{z - \varepsilon}{c} \right) \right]. \quad (9)$$

Theorem 1 (Smoothed-score homogeneity under oracle adaptive bandwidth). *Suppose (7) holds and $h(x) = cs(x)$. For any measurable score transform $A : [0, 1] \rightarrow \mathbb{R}$, define*

$$S_c^\circ(x, y) = A(F_c^\circ(y \mid x)).$$

Then

$$S_c^\circ(X, Y) \mid X = x \stackrel{d}{=} A(H_c(\varepsilon)).$$

Therefore the conditional distribution of the smoothed-oracle score does not depend on x .

Theorem 1 is the central scale-adaptive argument. Exact DCP with the true continuous CDF already has probability-integral-transform homogeneity. CKME-DCP uses smoothed indicators, and fixed smoothing can break this homogeneity through $h/s(x)$. Adaptive smoothing restores it at the smoothed-oracle level.

Assumption 3 (Uniform relative scale consistency). *The plug-in scale estimator satisfies*

$$\Delta_n = \sup_{x \in \mathcal{X}} \left| \frac{\widehat{s}(x)}{s(x)} - 1 \right| = o_{\mathbb{P}}(1).$$

Assumption 4 (Standardized moment bound). *For the response range $\mathcal{T}$,*

$$M_{\mathcal{T}} = \sup_{x \in \mathcal{X}, t \in \mathcal{T}} \mathbb{E} \left[\left| \frac{t - m(x)}{s(x)} - \varepsilon \right| \middle| X = x \right] < \infty.$$

The smooth step function Ψ is Lipschitz with constant L_{Ψ} .

Proposition 3 (Plug-in adaptive bandwidth is close to oracle adaptive bandwidth). *Suppose the location-scale model (7) holds and Assumptions 3–4 hold. Let*

$$F_h^\circ(t \mid x) = \mathbb{E} \left[\Psi \left(\frac{t - Y}{c\widehat{s}(x)} \right) \middle| X = x \right]$$

and let F_c° be the oracle adaptive smoothed CDF in (9). Then

$$\sup_{x \in \mathcal{X}, t \in \mathcal{T}} \left| F_h^\circ(t \mid x) - F_c^\circ(t \mid x) \right| = O_{\mathbb{P}}(\Delta_n) = o_{\mathbb{P}}(1).$$

5.3 CKME consistency and group-conditional behavior

Let the smoothed oracle target be

$$F_{h(\cdot)}^\circ(t \mid x) = \mathbb{E}[g_{t,h(x)}(Y) \mid X = x].$$

Assumption 5 (Uniform CKME embedding consistency). *For the chosen kernels and regularization sequence,*

$$\sup_{x \in \mathcal{X}} \|\widehat{\mu}_{Y|x} - \mu_{Y|x}\|_{\mathcal{H}_Y} = o_{\mathbb{P}}(1).$$

Assumption 6 (Bounded smoothed-indicator class). *For the target range $\mathcal{T}$,*

$$\sup_{x \in \mathcal{X}, t \in \mathcal{T}} \|g_{t,h(x)}\|_{\mathcal{H}_Y} < \infty.$$

Assumption 7 (Finite-partition oracle score regularity). *Let $\mathcal{G}_1, \dots, \mathcal{G}_K$ be a fixed finite partition of $\mathcal{X}$, with $\mathbb{P}(X \in \mathcal{G}_k) > 0$. For the oracle adaptive score $S_c^\circ(X, Y)$, the groupwise distribution functions*

$$G_k(q) = \mathbb{P}\{S_c^\circ(X, Y) \leq q \mid X \in \mathcal{G}_k\}$$

are Lipschitz in a common neighborhood of $q_{1-\alpha}^\circ$, uniformly over $k = 1, \dots, K$.

Proposition 4 (CKME embedding consistency implies smoothed-CDF consistency). *Under Assumptions 5–6,*

$$\sup_{x \in \mathcal{X}, t \in \mathcal{T}} \left| \widehat{F}_{h(\cdot)}(t \mid x) - F_{h(\cdot)}^\circ(t \mid x) \right| = o_{\mathbb{P}}(1).$$

Combining Proposition 4 with Proposition 2 gives score and threshold stability for CKME-DCP. Combining those with Theorem 1 yields the following group-conditional implication.

Theorem 2 (Asymptotic finite-partition group coverage). *Suppose the location-scale model holds, oracle adaptive smoothing $h(x) = cs(x)$ is used, and Assumptions 1, 2, 5, 6, and 7 hold with $F^\circ = F_c^\circ$. Then*

$$\max_{1 \leq k \leq K} \left| \mathbb{P}\{Y \in \widehat{C}(X) \mid X \in \mathcal{G}_k\} - (1 - \alpha) \right| \rightarrow 0$$

in probability.

Corollary 1 (Plug-in scale-adaptive CKME-DCP). *Suppose the location-scale model holds, the plug-in bandwidth $h(x) = c\widehat{s}(x)$ is used, and Assumptions 2, 3, 4, 5, 6, and 7 hold. If the CKME estimator is uniformly consistent for the plug-in smoothed oracle F_h° , then the conclusion of Theorem 2 continues to hold:*

$$\max_{1 \leq k \leq K} \left| \mathbb{P}\{Y \in \widehat{C}(X) \mid X \in \mathcal{G}_k\} - (1 - \alpha) \right| \rightarrow 0$$

in probability.

The theorem does not claim finite-sample pointwise conditional coverage, which is impossible in general without stronger assumptions. It states that, in the location-scale regime targeted here, adaptive smoothing plus uniform CKME CDF consistency can make DCP scores asymptotically homogeneous over fixed finite input partitions.

6 Positioning Relative to Existing CP Methods

The proposed method does not make conformal validity depend on CKME itself. Split conformal validity is estimator-agnostic once calibration and target samples are exchangeable under the target law. CKME enters through the score-construction layer, where a better conditional CDF estimator and a better smoothing scale can improve interval efficiency and groupwise behavior. The target-aware formulation therefore separates two roles:

target-aware calibration specifies where validity holds,
scale-adaptive CKME improves the DCP score used for efficiency.

Conformalized quantile regression adapts interval endpoints through estimated conditional quantiles [8]. Distribution regression versions of DCP estimate the conditional CDF through threshold-indexed regressions [4]. Localized and weighted conformal methods modify the calibration distribution or score weighting to improve local guarantees [12, 5]. CKME-DCP is complementary: it estimates a single conditional distributional object and uses smooth test functions to evaluate CDFs. This representation creates a bandwidth parameter whose local scaling can be matched to the response scale.

This distinction also clarifies the role of adaptive design. Adaptive site selection changes where simulation data are collected. It is a data-acquisition layer and can improve estimator quality in high-uncertainty regions, but it is not the same contribution as scale-adaptive smoothing. The present paper keeps the theorem line focused on target validity and scale-adaptive scoring.

7 Discussion

This paper separates the validity layer and the score-efficiency layer in CKME-DCP for stochastic simulation outputs. The validity layer is the split-conformal argument under a stated target joint law $P_{XY}^q = q_X P_{Y|X}$. The score-efficiency layer is the CKME conditional CDF estimator and, in particular, the smooth-indicator bandwidth used to evaluate DCP scores.

The target formulation makes the coverage statement explicit. Global-domain coverage, region-of-interest coverage, fixed-target-set average coverage, and smoothed fixed-target local coverage are different claims because they integrate over different input laws. The finite-sample split-conformal guarantee applies cleanly in the matched target-calibration setting, while target mismatch requires reweighting or a separate calibration design.

The scale-adaptive bandwidth addresses a different problem. Under heteroscedasticity, fixed smoothing creates the standardized ratio $h/s(x)$, so the same output-side bandwidth can represent different amounts of smoothing in low- and high-variance regions. Setting $h(x) = c\hat{s}(x)$ aligns the smoothed indicator with local response scale. The oracle analysis shows the mechanism.

A Proofs

A.1 Proof of Proposition 1

Condition on the training data and on the fitted estimator $\hat{F}$. Under the stated exchangeability condition, the calibration scores

$$S_i^{\text{cal}} = S_{\hat{F}}(X_i^{\text{cal}}, Y_i^{\text{cal}}), \quad i = 1, \dots, n_{\text{cal}},$$

and the target score

$$S_\star = S_{\widehat{F}}(X_\star, Y_\star)$$

are exchangeable. Let

$$k = \lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil.$$

If $k \leq n_{\text{cal}}$, $\widehat{q}_{1-\alpha}$ is the k -th order statistic of the calibration scores; if $k = n_{\text{cal}} + 1$, the usual convention is $\widehat{q}_{1-\alpha} = +\infty$, in which case the claim is trivial. For $k \leq n_{\text{cal}}$, the rank of $S_\star$ among the $n_{\text{cal}} + 1$ exchangeable scores is uniform up to ties, and the conservative tie convention gives

$$\mathbb{P}\{S_\star \leq \widehat{q}_{1-\alpha} \mid \mathcal{D}_{\text{tr}}, \widehat{F}\} \geq \frac{k}{n_{\text{cal}} + 1} \geq 1 - \alpha.$$

Since $Y_\star \in \widehat{C}(X_\star)$ is exactly the event $S_\star \leq \widehat{q}_{1-\alpha}$, taking expectation over the training data proves

$$\mathbb{P}_{(X,Y) \sim P_{XY}^q} \{Y \in \widehat{C}(X)\} \geq 1 - \alpha.$$

A.2 Proof of Proposition 2

Let $A(u) = |u - 1/2|$. This map is one-Lipschitz. On the event in Assumption 1,

$$\begin{aligned} |S_{\widehat{F}}(x, y) - S^\circ(x, y)| &= \left| A(\widehat{F}(y \mid x)) - A(F^\circ(y \mid x)) \right| \\ &\leq \left| \widehat{F}(y \mid x) - F^\circ(y \mid x) \right| \leq \epsilon_n \end{aligned}$$

uniformly over $x \in \mathcal{X}$ and $y \in \mathcal{T}$. This proves the score perturbation bound.

For the threshold statement, define the oracle calibration scores

$$S_i^\circ = S^\circ(X_i^{\text{cal}}, Y_i^{\text{cal}}), \quad i = 1, \dots, n_{\text{cal}},$$

and let $q_{n,1-\alpha}^\circ$ be their k -th order statistic, with the same $k = \lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil$. The uniform score perturbation bound implies

$$\max_{1 \leq i \leq n_{\text{cal}}} |\widehat{S}_i - S_i^\circ| \leq \epsilon_n.$$

Order statistics are one-Lipschitz with respect to the sup-norm perturbation of the sample, hence

$$|\widehat{q}_{1-\alpha} - q_{n,1-\alpha}^\circ| \leq \epsilon_n.$$

By Assumption 2, the oracle score distribution is regular at $q_{1-\alpha}^\circ$. Standard empirical-quantile theory gives

$$|q_{n,1-\alpha}^\circ - q_{1-\alpha}^\circ| = O_{\mathbb{P}}(n_{\text{cal}}^{-1/2}).$$

Combining the two displays yields

$$|\widehat{q}_{1-\alpha} - q_{1-\alpha}^\circ| = O_{\mathbb{P}}(\epsilon_n + n_{\text{cal}}^{-1/2}).$$

A.3 Proof of Theorem 1

Under $Y = m(x) + s(x)\varepsilon$ and $h(x) = cs(x)$,

$$F_c^\circ(y \mid x) = H_c\left(\frac{y - m(x)}{s(x)}\right).$$

Evaluating at $Y = m(x) + s(x)\varepsilon$ gives

$$F_c^\circ(Y \mid X = x) = H_c(\varepsilon).$$

Because $\varepsilon \perp X$, the conditional distribution of $H_c(\varepsilon)$ given $X = x$ is the same for all x . Applying any measurable transform A preserves equality in distribution, so

$$S_c^\circ(X, Y) \mid X = x = A(F_c^\circ(Y \mid X = x)) \stackrel{d}{=} A(H_c(\varepsilon)).$$

A.4 Proof of Proposition 3

Write

$$z_x(t) = \frac{t - m(x)}{s(x)}.$$

Under the location-scale model,

$$\frac{t - Y}{c\hat{s}(x)} = \frac{z_x(t) - \varepsilon s(x)}{c \hat{s}(x)}, \quad \frac{t - Y}{cs(x)} = \frac{z_x(t) - \varepsilon}{c}.$$

On the event $\Delta_n < 1/2$,

$$\sup_{x \in \mathcal{X}} \left| \frac{s(x)}{\hat{s}(x)} - 1 \right| = \sup_{x \in \mathcal{X}} \left| \frac{1}{\hat{s}(x)/s(x)} - 1 \right| \leq 2\Delta_n.$$

Using the Lipschitz property of Ψ , for every x, t ,

$$\begin{aligned} & \left| F_h^\circ(t \mid x) - F_c^\circ(t \mid x) \right| \\ & \leq \mathbb{E} \left[L_\Psi \left| \frac{z_x(t) - \varepsilon}{c} \right| \left| \frac{s(x)}{\hat{s}(x)} - 1 \right| \mid X = x \right] \\ & \leq \frac{2L_\Psi}{c} \Delta_n \mathbb{E} [|z_x(t) - \varepsilon| \mid X = x]. \end{aligned}$$

Taking the supremum over $x \in \mathcal{X}$ and $t \in \mathcal{T}$, and applying Assumption 4, gives

$$\sup_{x, t} \left| F_h^\circ(t \mid x) - F_c^\circ(t \mid x) \right| \leq \frac{2L_\Psi M_{\mathcal{T}}}{c} \Delta_n$$

with probability tending to one. Assumption 3 gives the result.

A.5 Proof of Proposition 4

For every x, t ,

$$\left| \hat{F}_{h(\cdot)}(t \mid x) - F_{h(\cdot)}^\circ(t \mid x) \right| = \left| \langle g_{t, h(x)}, \hat{\mu}_{Y|x} - \mu_{Y|x} \rangle_{\mathcal{H}_Y} \right| \leq \|g_{t, h(x)}\|_{\mathcal{H}_Y} \|\hat{\mu}_{Y|x} - \mu_{Y|x}\|_{\mathcal{H}_Y}.$$

Taking suprema and applying Assumptions 5–6 gives the result.

A.6 Proof of Theorem 2

Let

$$\widehat{S}(x, y) = S_{\widehat{F}}(x, y), \quad S_c^\circ(x, y) = |F_c^\circ(y \mid x) - 1/2|.$$

By Proposition 4 and Proposition 2, there exists a sequence $r_n = o_{\mathbb{P}}(1)$ such that, with probability tending to one,

$$\sup_{x, y} \left| \widehat{S}(x, y) - S_c^\circ(x, y) \right| \leq r_n, \quad \left| \widehat{q}_{1-\alpha} - q_{1-\alpha}^\circ \right| \leq r_n.$$

On this event, for any group $\mathcal{G}_k$,

$$\{S_c^\circ(X, Y) \leq q_{1-\alpha}^\circ - 2r_n\} \subseteq \{\widehat{S}(X, Y) \leq \widehat{q}_{1-\alpha}\} \subseteq \{S_c^\circ(X, Y) \leq q_{1-\alpha}^\circ + 2r_n\}.$$

Therefore,

$$\begin{aligned} & \left| \mathbb{P}\{\widehat{S}(X, Y) \leq \widehat{q}_{1-\alpha} \mid X \in \mathcal{G}_k\} - \mathbb{P}\{S_c^\circ(X, Y) \leq q_{1-\alpha}^\circ \mid X \in \mathcal{G}_k\} \right| \\ & \leq \sup_{|u - q_{1-\alpha}^\circ| \leq 2r_n} |G_k(u) - G_k(q_{1-\alpha}^\circ)|, \end{aligned}$$

where G_k is the groupwise oracle score CDF in Assumption 7. By the uniform Lipschitz condition, the right-hand side is $O(r_n) = o_{\mathbb{P}}(1)$, uniformly over the fixed finite set of groups.

It remains to identify the oracle group coverage level. Theorem 1 implies that the conditional distribution of $S_c^\circ(X, Y)$ given $X = x$ does not depend on x . Hence every group has the same oracle score distribution as the marginal target distribution:

$$\mathbb{P}\{S_c^\circ(X, Y) \leq u \mid X \in \mathcal{G}_k\} = \mathbb{P}\{S_c^\circ(X, Y) \leq u\}.$$

At the continuity point $q_{1-\alpha}^\circ$, this probability equals $1 - \alpha$. Thus

$$\max_{1 \leq k \leq K} \left| \mathbb{P}\{Y \in \widehat{C}(X) \mid X \in \mathcal{G}_k\} - (1 - \alpha) \right| \rightarrow 0$$

in probability.

A.7 Proof of Corollary 1

Let $F_{\widehat{h}}^\circ$ denote the plug-in smoothed oracle CDF and let F_c° denote the oracle adaptive smoothed CDF. By the assumed uniform CKME consistency for $F_{\widehat{h}}^\circ$,

$$\sup_{x, t} \left| \widehat{F}_{\widehat{h}(\cdot)}(t \mid x) - F_{\widehat{h}}^\circ(t \mid x) \right| = o_{\mathbb{P}}(1).$$

By Proposition 3,

$$\sup_{x, t} \left| F_{\widehat{h}}^\circ(t \mid x) - F_c^\circ(t \mid x) \right| = o_{\mathbb{P}}(1).$$

The triangle inequality gives

$$\sup_{x, t} \left| \widehat{F}_{\widehat{h}(\cdot)}(t \mid x) - F_c^\circ(t \mid x) \right| = o_{\mathbb{P}}(1).$$

Thus Assumption 1 holds with $F^\circ = F_c^\circ$, up to replacing ϵ_n by the sum of the CKME and plug-in scale errors. The proof of Theorem 2 then applies verbatim.

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